

IMO Training – Number Theory

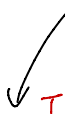
Infinite Descent and Vieta Jumping

無窮下降法

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Notes

Idea for  Fermat's Infinite Descent

- Aim to prove "No solution".
- Assume the contrary, i.e. $\exists$ some solution
- Pick the smallest one
- After some Maths, I can find an even smaller one.
- Contradicts MINIMALITY of the solution you picked.

① **Example.** Find all positive integral solutions to the system $\begin{cases} a^2 + 6b^2 = p^2 & \text{--- ①} \\ b^2 + 6a^2 = q^2 & \text{--- ②} \end{cases}$

② **Example.** Show that $\sqrt[3]{2}$ is irrational.

② Assume the contrary, i.e. $\exists a, b \in \mathbb{N}$ s.t. $\sqrt[3]{2} = \frac{a}{b}$.

So $a^3 = 2b^3$. Choose ^{such} minimal a, b ,
which is impossible by F.L.T. $\#$
 $\therefore \sqrt[3]{2}$ irrational

Then a even, so let $a = 2a_1$,

some $a_1 \in \mathbb{N}$. So $8a_1^3 = 2b^3$,

$$4a_1^3 = b^3.$$

Then b even, so let $b = 2b_1$, some $b_1 \in \mathbb{N}$.

$$\therefore 4a_1^3 = 8b_1^3$$

$$a_1^3 = 2b_1^3.$$

So (a_1, b_1) is another pair of natural numbers.

s.t. $\frac{a_1}{b_1} = \sqrt[3]{2}$, but it contradicts the minimality of (a, b) . $\#$

$\therefore \sqrt[3]{2}$ irrational.

① Consider the minimal solution (a, b, p, q) .

$$\text{①+②, } 7a^2 + 7b^2 = p^2 + q^2.$$

$$p^2 + q^2 \equiv 0(7).$$

$$p^2 \equiv -q^2(7).$$

$$\therefore p^2 \equiv 0, 1, 2, 4(7).$$

$$\therefore \text{For } p^2 \equiv -q^2(7),$$

$$p^2, q^2 \equiv 0(7).$$

$$7 \mid p, q.$$

Let $p = 7p_1, q = 7q_1$,
some $p_1, q_1 \in \mathbb{N}$.

Then the original equation becomes

$$7a^2 + 7b^2 = 49p_1^2 + 49q_1^2$$

$$a^2 + b^2 = 7p_1^2 + 7q_1^2.$$

Then we have a smaller solution (p_1, q_1, a, b) , which contradicts minimality. $\#$

$\therefore$ No solution.

① **Example.** Solve $x^3 + 2y^3 = 4z^3$ in non-negative integers.

② **Example.** Prove that $x^4 + y^4 = z^4$ has no positive integral solution.

Corollary. For all positive integers k , prove that $x^{4k} + y^{4k} = z^{4k}$ has no positive integral solution.

Consider minimal solution $(x, y, z) \in \mathbb{N}^3$.

① x is even. So let $x = 2x_1$, some $x_1 \in \mathbb{N}$.

Then the eqⁿ becomes $8x_1^3 + 2y^3 = 4z^3$

$$y^3 + 4x_1^3 = 2z^3.$$

y is even. So let $y = 2y_1$, some $y_1 \in \mathbb{N}$.

$$8y_1^3 + 4x_1^3 = 2z^3.$$

$$4y_1^3 + 2x_1^3 = z^3$$

z is even. So let $z = 2z_1$, some $z_1 \in \mathbb{N}$.

$$4y_1^3 + 2x_1^3 = 8z_1^3$$

$$x_1^3 + 2y_1^3 = 4z_1^3.$$

$\therefore (x_1, y_1, z_1)$ is a smaller solution set,

which contradicts the minimality.

$\therefore (x, y, z) = (0, 0, 0)$ is the only possible solⁿ,

$$\text{Indeed, } 0^3 + 2 \cdot 0^3 = 4 \cdot 0^3.$$

$$= 0.$$

Cor. If $\exists x, y, z \in \mathbb{N}$ s.t. $x^{4k} + y^{4k} = z^{4k}$,

$$(x^k, y^k, z^k) \text{ then } (x^k)^4 + (y^k)^4 = (z^k)^4$$

is a solution to the equation.

$$a^4 + b^4 = c^4, \text{ which is}$$

impossible.

□₃

② Consider minimal solution (x, y, z) to the eqⁿ

$$(x^2)^2 + (y^2)^2 = (z^2)^2,$$

(x^2, y^2, z^2) is a Pythagorean triple.

$\therefore \exists$ coprime $p, q \in \mathbb{N}$ s.t.

$$\begin{cases} x^2 = 2pq \\ y^2 = p^2 - q^2 \\ z^2 = p^2 + q^2 \end{cases} \leftarrow p^2 = y^2 + q^2.$$

In particular, $\exists$ coprime $a, b \in \mathbb{N}$ s.t.

$$\begin{cases} q = 2ab \\ y = a^2 - b^2 \\ p = a^2 + b^2 \end{cases}$$

$$\text{Then } x^2 = 4ab(a^2 + b^2).$$

$\therefore a, b, a^2 + b^2$ coprime.

$\therefore a, b, a^2 + b^2$ are all perfect squares.

Let $a = A^2, b = B^2, a^2 + b^2 = p^2$, some $A, B, p \in \mathbb{N}$. But then

$$A^4 + B^4 = p^2, \text{ which contradicts minimality of } x, y, z$$

$\therefore$ By Infinite Descent, the eqⁿ

$$x^4 + y^4 = z^2 \text{ has no solution}$$

In particular, $x^4 + y^4 = z^4$ has no solution.

① **Example.** (Putnam 1988) Prove that $\forall n \in \mathbb{R}$, the equation

$$x_1^2 + x_2^2 + x_3^2 + x_4^2 = x_1x_2x_3 + x_1x_2x_4 + x_1x_3x_4 + x_2x_3x_4$$

has a solution for which x_1, x_2, x_3, x_4 are all integers larger than n .

② **Example.** (IMO 2007) Let $a, b \in \mathbb{N}$. Show that if $4ab - 1 \mid (4a^2 - 1)^2$, then $a = b$.

① $(x_1, x_2, x_3, x_4) = (1, 1, 1, 1)$ is an obvious solution.

Rewrite the equation as

$$x_1^2 - (x_2x_3 + x_2x_4 + x_3x_4)x_1 + x_2^2 + x_3^2 + x_4^2 - x_2x_3x_4 = 0.$$

1 1 1 1
1 1 1 2
4 1 1 2
1 1 2 4
13 1 2 4
1 2 4 13
85 2 4 13

2 4 13 85

Let $x_1 = r_1, r_2$ be the roots to the above equation,

$$\text{then } r_1 + r_2 = x_2x_3 + x_2x_4 + x_3x_4.$$

If $(x_1, x_2, x_3, x_4) = (a, b, c, d)$, and wlog, $a \leq b \leq c \leq d$.
Then $(x_1, x_2, x_3, x_4) = (bc + cd + bd - a, b, c, d)$ is another solution set.

$\therefore bc + cd + bd - a \geq d \geq c \geq b$, so we can generate a larger

solution.

$\therefore$ We have arbitrarily large solutions, as desired.

Example. (IMO 2007) Let $a, b \in \mathbb{N}$. Show that if $4ab - 1 \mid (4a^2 - 1)^2$, then $a = b$.

$$4ab - 1 \mid (4a^2 - 1)^2 b^2$$

$$16a^2 b^2 = (4ab)^2$$

$$\equiv 1^2 \pmod{4ab-1}$$

$$\equiv 1 \pmod{4ab-1}$$

$$\Leftrightarrow 4ab - 1 \mid 16a^4 b^2 - 8a^2 b^2 + b^2$$

$$\Leftrightarrow 4ab - 1 \mid (16a^2 b^2) a^2 - (4ab)(2ab) + b^2$$

$$\Leftrightarrow 4ab - 1 \mid a^2 - 2ab + b^2$$

$$\Leftrightarrow 4ab - 1 \mid (a - b)^2$$

Let (a_1, b_1) be a solution set, and wlog, let $a_1 < b_1$.

Then let $k = \frac{(a - b_1)^2}{4ab_1 - 1}$. Reunitiy.

Pick the set with $a_1 + b_1$ minimum.

$$a^2 - 2ab_1 + b_1^2 = k(4ab_1 - 1)$$

$$a^2 - a(2b_1 + 4b_1 k) + b_1^2 + k = 0$$

Let $a = a_1, a_2$ be the roots to the above equation.

$$\text{Then } \begin{cases} a_1 + a_2 = 2b_1 + 4b_1 k & (1) \\ a_1 a_2 = b_1^2 + k & (2) \end{cases}$$

From (1), $a_2 \in \mathbb{Z}$.

From (2), $a_2, b_1, k \in \mathbb{N}$, $a_2 \in \mathbb{N}$.

Now $a_2 < a_1 \Leftrightarrow \frac{b_1^2 + k}{a_1} < a_1$

(From (2))

$$\Leftrightarrow b_1^2 + k < a_1^2$$

$$\Leftrightarrow b_1^2 + \frac{(a_1 - b_1)^2}{4a_1 b_1 - 1} < a_1^2$$

$$\Leftrightarrow \frac{(a_1 - b_1)^2}{4a_1 b_1 - 1} < a_1^2 - b_1^2$$

$$\Leftrightarrow \frac{a_1 - b_1}{4a_1 b_1 - 1} < a_1 + b_1$$

which is true because

$$\frac{a_1 - b_1}{4a_1 b_1 - 1} < a_1 - b_1 < a_1 + b_1$$

But this contradicts the minimality of $a_1 + b_1$.

Similar result will occur when I assume $a_1 > b_1$. $\therefore a = b$

① **Example.** Let $x, y \in \mathbb{N}$ such that $xy \mid x^2 + y^2 + 1$. Prove that $\frac{x^2 + y^2 + 1}{xy} = 3$.

② **Example.** Let $a, b \in \mathbb{N}$ with $ab \neq 1$. Suppose that $ab - 1 \mid a^2 + b^2$. Show that $\frac{a^2 + b^2}{ab - 1} = 5$.

Let (x_1, y_1) be a solution with $x_1 + y_1$ minimal. Let $\frac{x_1^2 + y_1^2 + 1}{x_1 y_1} = k \in \mathbb{N}$.

Wolog, $x_1 \geq y_1$. If $x_1 = y_1$, then $x_1^2 \mid 1$, so $x_1 = 1$, $y_1 = 1$, $k = 3$.

Otherwise, $x_1 > y_1$. Then, ^{consider} $x^2 + y_1^2 + 1 = k x y_1$.

$$x^2 - k x y_1 + y_1^2 + 1 = 0.$$

Let x_1, x_2 be the roots to the above eqⁿ, so $\begin{cases} x_1 + x_2 = k y_1 \\ x_1 x_2 = y_1^2 + 1 \end{cases}$.

$\therefore x_2 = k y_1 - x_1$, $\therefore x_2 \in \mathbb{Z}$.

$\therefore x_1 x_2 > 0$ and $x_1 > 0$, so $x_2 \in \mathbb{N}$.

Now $x_2 = \frac{y_1^2 + 1}{x_1}$. Note that $x_2 < x_1$
 $\Leftrightarrow \frac{y_1^2 + 1}{x_1} < x_1$

$$\Leftrightarrow y_1^2 + 1 < x_1^2.$$

which is true as $x_1^2 \geq (y_1 + 1)^2 > y_1^2 + 1$.

But this contradicts the minimality of $x_1 + y_1$.

$\therefore x_1 = y_1$, $k = 3$.

□

② Example. Let $a, b \in \mathbb{N}$ with $ab \neq 1$. Suppose that $ab - 1 \mid a^2 + b^2$. Show that $\frac{a^2 + b^2}{ab - 1} = 5$.

Let (a_1, b_1) be a solution with $a_1 + b_1$ minimal. wlog, $a_1 \geq b_1$.

$$\text{Let } \frac{a_1^2 + b_1^2}{a_1 b_1 - 1} = k. \text{ If } a_1 = b_1, \text{ then } k = \frac{2a_1^2}{a_1^2 - 1} = 2 + \frac{2}{a_1^2 - 1},$$

then $a_1^2 - 1 = 1$ or 2 . No solution. So $a_1 \neq b_1$.

wlog, $a_1 \geq b_1 + 1$. For $\frac{a_1^2 + b_1^2}{a_1 b_1 - 1} = k$,

$$a_1^2 + b_1^2 = k a_1 b_1 - k$$

$$a_1^2 - (k b_1) a_1 + b_1^2 + k = 0.$$

Let $a = a_1, a_2$ be the roots to the above eqⁿ.

$$\text{Then } \begin{cases} a_1 + a_2 = k b_1 \\ a_1 a_2 = b_1^2 + k. \end{cases}$$

$$\therefore a_2 = k b_1 - a_1, \therefore a_2 \in \mathbb{Z}.$$

$$\therefore a_1 > 0 \text{ \& } b_1^2 + k > 0, \therefore a_2 \in \mathbb{N}. \text{ Notice that}$$

$$a_2 < a_1$$

$$\Leftrightarrow \frac{b_1^2 + k}{a_1} < a_1$$

$$\Leftrightarrow b_1^2 + k < a_1^2.$$

$$\Leftrightarrow \frac{a_1^2 + b_1^2}{a_1 b_1 - 1} < a_1^2 - b_1^2$$

$$\Leftrightarrow a_1^2 + b_1^2 < (a_1 b_1 - 1)(a_1 + b_1)(a_1 - b_1).$$

$$\Leftrightarrow a_1^2 + b_1^2 < (a_1^2 b_1 - a_1 + a_1 b_1^2 - b_1)(a_1 - b_1)$$

$$\Leftrightarrow a_1^2 + b_1^2 < (a_1(a_1 b_1 - 1) + b_1(a_1 b_1 - 1))(a_1 - b_1).$$

If $a_1, b_1 \geq 2$, the ineq. is true, no solution.

otherwise, a_1 or $b_1 = 1$. But since $a_1 \neq b_1$, and $a_1 > b_1$,

$$\text{take } b_1 = 1. \text{ Then } k = \frac{a_1^2 + 1}{a_1 - 1} = a_1 + 1 + \frac{2}{a_1 - 1}.$$

$$\text{So } a_1 - 1 = 1 \text{ or } 2, \quad a_1 = 2, 3.$$

$$\text{But when } a_1 = 2 \text{ or } 3, \quad k = \frac{2^2 + 1}{2 - 1} = 5 \sim k = \frac{3^2 + 1}{3 - 1} = 5.$$

$\therefore$ Only possible solution is $k = 5$.

Example. (Crux) If $a, b, c \in \mathbb{N}$ such that $0 < a^2 + b^2 - abc \leq c$, show that $a^2 + b^2 - abc$ is a perfect square.

Example. (IMO 1988 Problem 6) Let $a, b \in \mathbb{N}$ such that $ab + 1 \mid a^2 + b^2$. Prove that $\frac{a^2 + b^2}{ab + 1}$ is a perfect square.

1988 Problem 6.

Let $\frac{a^2 + b^2}{ab + 1} = k \in \mathbb{N}$. Assume k is not a perfect square.

Let (a_1, b_1) is a solution to the above equation with $a_1 + b_1$ minimal.

Then $a_1^2 + b_1^2 = k(a_1 b_1 + 1)$.

$$a_1^2 - (k b_1) a_1 + b_1^2 - k = 0.$$

Consider the equation $x^2 - k b_1 x + b_1^2 - k = 0$.

Let $x = a_1, a_2$ be the roots to the equation. By Vieta,

$$\begin{cases} a_1 + a_2 = k b_1 & \text{--- (1)} \\ a_1 a_2 = b_1^2 - k & \text{--- (2)} \end{cases}$$

From (1), $a_2 = k b_1 - a_1$, so $a_2 \in \mathbb{Z}$. Assume $a_2 < 0$, then

$$k = \frac{a_1^2 + b_1^2}{a_1 b_1 + 1} = \frac{a_1^2 - k a_2 b_1 + b_1^2}{a_1 b_1 + 1}$$

$$\geq \frac{a_1^2 + k b_1 + b_1^2}{a_1 b_1 + 1}$$

$$> k. \quad \#$$

$\therefore k$ is not a perfect square.
(contradiction),

$$\therefore b_1^2 - k \neq 0, \quad a_2 \neq 0.$$

$$\therefore a_2 \in \mathbb{N}.$$

$$\text{Now } a_2 < a_1 \iff \frac{b_1^2 - k}{a_1} < a_1$$

$$\iff b_1^2 - k < a_1^2$$

which is true as $b_1 \leq a_1$

But this contradicts the minimality of $a_1 + b_1$. $\#$

$\therefore \frac{a^2 + b^2}{ab + 1}$ is a perfect square. $\square$

Extra Space

Assume that $\exists a, b, c \in \mathbb{N}$ s.t. $a^2 + b^2 - abc = k$ is not a perfect square.

Also, $k > 0$ & $k \leq c$. Consider (a, b) satisfying the above equation, with $a+b$ minimal. WLOG, $a \geq b$.

Consider equation $x^2 - (bc)x + b^2 - k = 0$.

Let $x = a_1, a_2$ be the roots to the above equation.

By Vieta, $\begin{cases} a_1 + a_2 = bc & (1) \\ a_1 a_2 = b^2 - k & (2) \end{cases}$

From (1), $a_2 = bc - a_1$, so $a_2 \in \mathbb{Z}$. assumption

If $a_1 = 0$, then $k = b^2$, contradicting the $\neg$ that k is not a perfect square. If $a_2 < 0$,

$$\begin{aligned} k &= a_2^2 - bc \cdot a_2 + b^2 \\ &\geq a_2^2 + bc + b^2 \\ &> c \end{aligned} \quad \therefore a_2 \notin \mathbb{N}.$$

which contradicts $k \leq c$. ~~###~~

Now $a_2 < a_1$

$$\Leftrightarrow \frac{b^2 - k}{a_1} < a_1$$

$$b^2 - k < a_1^2$$

which is true because $b < a_1$. γ

But this contradicts the minimality of $a+b$. ~~###~~

$\therefore a^2 + b^2 - abc$ is a perfect square.